

T-030 / R-041 (R-033 step 1): Sidon-height latitude unions obey

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

first issued 2026-06-14 · this version issued 2026-06-14 · v1.1

1. Purpose and scope

Operator direction: after the sphere L -dependence (R-040), proceed to R-033 – bound the off-diagonal four-circle in-plane incidence energy below $N^{9/4}$, which would lift the sphere $E_+ = O(N^2)$ conjecture (R-040) to a theorem. This note is step 1: it PROVES the incidence-FREE part (Sidon heights) and pins the residual exactly. Scope: parallel circles (latitudes); the H-NONLATTICE-REMAINDER frontier.

2. Theorem: Sidon heights give $E_+ \leq 6N^2$ (any radii)

Theorem 1 (Sidon-height bound). *Let $Q = \bigcup_{i=1}^L (C_i \times \{z_i\})$ be a union of circles in parallel planes at heights z_i (ANY radii ρ_i), $N = |Q|$, with the heights $\{z_i\}$ a SIDON set (i.e. $z_a + z_b = z_c + z_d \Rightarrow \{z_a, z_b\} = \{z_c, z_d\}$). Then*

$$E_+(Q) \leq 6N^2.$$

Proof. In $E_+ = \#\{(a, b, c, d) : a + b = c + d\}$ the height coordinate gives $z_a + z_b = z_c + z_d$, so by the Sidon property the height multiset matches, $\{z_a, z_b\} = \{z_c, z_d\}$, hence the plane-index multiset $\{i_a, i_b\} = \{i_c, i_d\}$. Indexing by the ordered pair (i_a, i_b) , the matched (i_c, i_d) is one of $(i_a, i_b), (i_b, i_a)$ – at most two arrangements. The in-plane count of each is either the rectangular energy $E_+(C_i, C_j)$ (arrangement (i, j, i, j)) or the difference-correlation $\text{corr}(C_i, C_j) = \sum_v d_{C_i}(v) d_{C_j}(v)$ (arrangement (i, j, j, i)); both are $\leq 3\nu_i \nu_j$ – the first by R-021 (E_+ of two circles), the second by Cauchy–Schwarz with $E_+(C) \leq 3\nu^2$. Summing over ordered (i, j) ,

$$E_+(Q) \leq \sum_{i,j} [E_+(C_i, C_j) + \text{corr}(C_i, C_j)] \leq 6 \sum_{i,j} \nu_i \nu_j = 6N^2. \quad \square$$

The bound uses the Sidon property of the HEIGHTS only; the radii are arbitrary (machine: assert A7, equal radius 25 + Sidon heights still gives $E_+ \leq 6N^2$). Measured $E_+/N^2 \leq 2.6$ on the tested families (assert A2); the proved constant 6 allows for both arrangements. The Sidon-decoupling step is verified exactly: zero quadruples with $a + b = c + d$ but mismatched height pair (assert A1).

3. The residual is the open four-circle incidence energy (R-033)

For NON-Sidon heights (e.g. an arithmetic progression) the decoupling fails: there are $a + b = c + d$ with $\{z_a, z_b\} \neq \{z_c, z_d\}$ (assert A4: 1248 such quadruples at $L = 5$, AP heights). These mismatched-height quadruples are the off-diagonal four-circle in-plane incidences

$$\#\{c_a + c_b = c_c + c_d : c_a \in C_i, c_b \in C_j, c_c \in C_k, c_d \in C_l\}, \quad \{i, j\} \neq \{k, l\},$$

and they do NOT reduce to a single bounded 2-D concentric-circle energy: that 2-D energy GROWS with L (assert A5: $E_+^{2D}/N^2 = 3.2, 3.9, 5.4, 6.0$ for $L = 2, 4, 6, 8$), because the 3-D heights suppress

collisions that the 2-D projection overcounts. Hence the residual is genuinely height-aware: bounding it by $O(N^2 \log^{O(1)} N)$ (or below the current $N^{9/4}$, R-033) is exactly the open circle-incidence problem, and closing it lifts the R-040 sphere conjecture to a theorem. **This is the precise next target (R-033).**

4. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

- (α) “The constant is $3N^2$, not $6N^2$.” CORRECTED in this note: an intermediate draft used $3N^2$ via a single-arrangement identity; the verification decomposition shows BOTH (i, j, i, j) and (i, j, j, i) contribute, so the proved constant is 6 (assert A3: $E_+ \leq \sum[E_+(C_i, C_j) + \text{corr}] \leq 6N^2$). The measured value $\leq 2.6N^2$ is well inside.
- (β) “The residual reduces to the 2-D concentric energy.” REFUTED by the data and recorded: the 2-D concentric energy grows with L (assert A5), so it is an overcount; the genuine residual keeps the height structure. No false reduction is claimed.
- (γ) “Does this need distinct radii like R-040?” NO: the Sidon decoupling is a height property; assert A7 confirms equal radii + Sidon heights still give $E_+ \leq 6N^2$. This is strictly cleaner than R-040’s distinct-radius requirement.
- (δ) “Is this a closure of T-030?” No: only the Sidon-height case is closed (incidence-free); the general (non-Sidon) case is the open four-circle incidence energy. T-030 stays OPEN; B5 unchanged.

Quantitative sanity check (mandatory). *Decoupling.* 0 mismatched quadruples for Sidon heights (assert A1) – the proof’s load-bearing step. *Bound.* $E_+ \leq 6N^2$ on all Sidon configs to $L = 8$ (assert A2), and $E_+ \leq \sum[E_+(C_i, C_j) + \text{corr}] \leq 6N^2$ exactly at $L = 5$ (assert A3). *Radius-agnostic.* equal radius + Sidon $\Rightarrow E_+/N^2 = \text{measured} \leq \text{bound}$ (assert A7). *Residual nonzero.* 1248 mismatched quadruples at $L = 5$ AP (assert A4). *Reproducibility.* deterministic exact-integer; 7/7 asserts, exit 0.

5. Result footer

Result ID:	R-041 (Sidon-height bound; R-033 step 1) / T-030 frontier
Precise statement:	L latitude circles (parallel planes, ANY radii) at heights z_i forming a SIDON set $\Rightarrow E_+(Q) \leq 6N^2$, incidence-free (height decoupling: $\{z_a, z_b\} = \{z_c, z_d\}$; two arrangements $(i, j, i, j), (i, j, j, i)$, each $\leq 3 \nu_i \nu_j$ by R-021 / Cauchy-Schwarz). RADIUS-AGNOSTIC. Non-Sidon residual = off-diagonal four-circle in-plane incidence energy (does NOT reduce to a bounded 2-D concentric energy, which grows with L); that is the open R-033 core.
Scope:	parallel circles (latitudes); T-030 (Q in S^2 ; H-NONLATTICE-REMAINDER). Not load-bearing for C_full head.
Evidence grade:	PROVED (T7-within-class: Sidon heights, any radii). Isolates the incidence-free part of the R-040 sphere $O(N^2)$ conjecture (which stays STRONG EVIDENCE T4). Does NOT close T-030: arbitrary- Q $N^{\{2+\epsilon\}}$ OPEN ($N^{\{9/4\}}$, R-033).
Reproduction command:	python3 codes/vacuum/dr2_t030_sidon_decoupling.py
Expected output:	7/7 asserts PASS, exit 0; JSON at claims/B5-BEYOND-LAYER-BOUND/runs/260614-dr2-t030-sidon-decoupling/result.json
Falsifier:	a Sidon-height latitude union with $E_+ > 6N^2$ (refutes the theorem); a Sidon-height config with a mismatched-height quadruple (refutes the decoupling). Both machine-checked false.

Tier before / after: NONE -- operator decides. No gate/tier/hypothesis flip.
 B5 stays T7-SCOPE_{admissibility-bounded} given A1;
 T-030 OPEN.

No-overclaim statement: Only the Sidon-height case is proved (incidence-free,
 constant 6). The general (non-Sidon) case is the open
 four-circle incidence energy; the 2-D concentric
 reduction was refuted (it grows with L) and is not
 claimed. Builds on R-021/R-025/R-040.

Next required action: R-033: bound the off-diagonal four-circle in-plane
 incidence energy by $O(N^2 \text{ polylog})$ or below $N^{9/4}$ ->
 lifts the sphere $O(N^2)$ conjecture to a theorem.